ENEL 585 – Remote HIL Quick Start

Running Your Controller on the Remote HIL: Orientation Lab and Final Project

PLECS will not run on the remote HIL computer: its license blocks Remote Desktop (FlexNet error -103), so External Mode is unavailable. Instead, build the `.out` with PLECS on your own computer, then flash it and change parameters live from the CCS **Watch** panel. Practise with the orientation lab model; your project model works the same way.

New tool: Code Composer Studio (CCS). CCS is TI's development environment for its microcontrollers, and it is installed on the HIL computers. You use only its debugger, to load the `.out` onto the LaunchPad, start it, and read or change variables while it runs. You write no code and create no CCS project; the steps below are all you need.

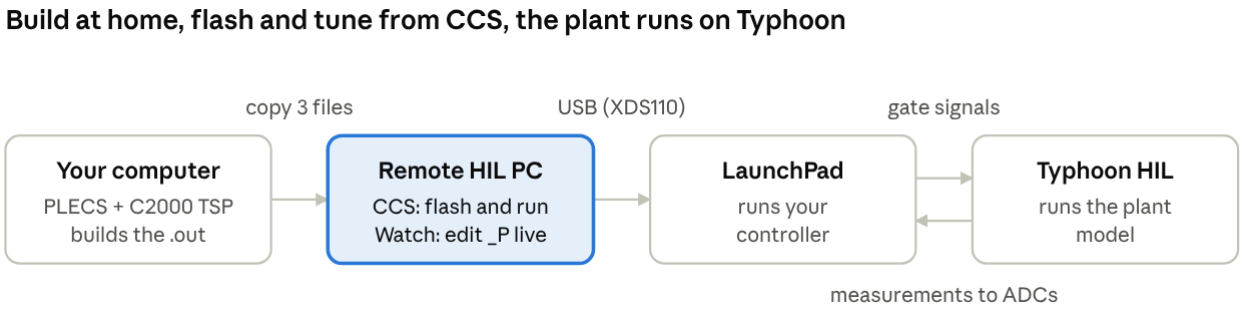


Fig. 1 – Remote HIL workflow. Only the `.out` file and two helper files cross from your computer to the HIL computer; PLECS never runs there.

1. Build on Your Computer

In **Coder → Coder options… → Target**, set **Build type** to **Build only** and keep the other target settings. Click **Build**, then copy these files from `<model>_codegen` to the HIL computer:

- `output_TI28P55x/<name>.out`: the firmware
- `Launchpad_TMS320F28P550SJ9.ccxml`: the CCS connection settings (also on D2L; it is the same for every model)
- `<name>.h`: lists the variable names you will need in Watch

`<name>` is the name of the part PLECS generates code for: the model name for the orientation model, and `Controller` for the project models (`Controller.out`, `Controller.h`).

2. Flash and Run (CCS 20 on the HIL Computer)

1. Start the plant model in Typhoon HIL. Orientation: as in the Lab 0 instructions. Project: open your project's `.tse` from D2L, compile it, then open its `.cus` SCADA panel and start the simulation.
2. In CCS, **File → Open Folder…** and open the folder with your files. If the menu is hidden, click the menu icon (three lines) at the top left.
3. Right-click the `.ccxml` → **Start Project-less Debug**.
4. In **Run and Debug**, select **C28xx_CPU1**. If it shows Disconnected, right-click → **Connect Target**.
5. **Run → Load Program** → your `.out`. This writes it to flash.
6. Press **Continue** (▶ or F5).

3. Change Parameters Live

Tunable parameters are in `<name>_P` and act on the next control step. Computed signals are in `<name>_B`: watch them only, since every step overwrites them.

A variable name labels one value in the LaunchPad's memory. PLECS builds it from the code and block names: `<name>_P.Vout_ref_Value` is the Value of the Constant block `Vout_ref`. You change the value, not the name; to get a different name, rename the block in PLECS and rebuild.

In **Run and Debug → Watch**, click **+**, enter `<name>_P` as the expression and click **OK**. Expand it to see its fields, then double-click a value to edit it.

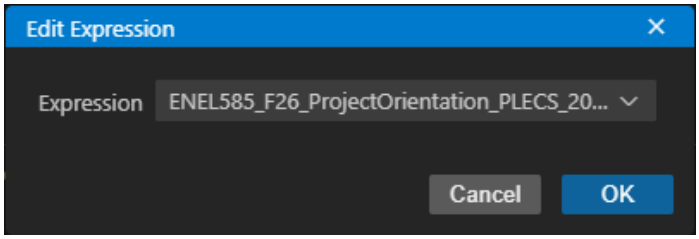


Fig. 2 – Adding the parameter structure to Watch.

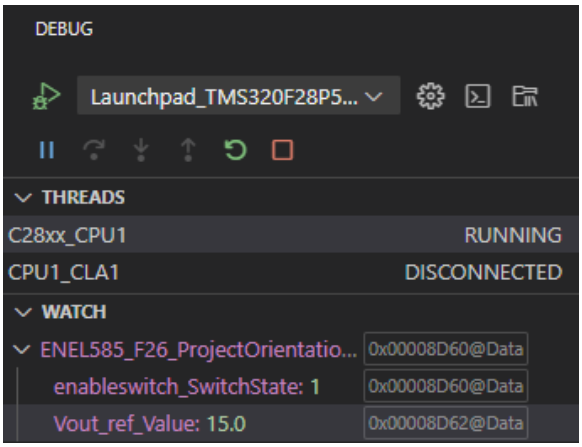


Fig. 3 – Run and Debug view with the program running: switching enabled and `Vout_ref_Value` changed to 15. CPU1_CLA1 stays disconnected; that is normal.

Prefix each field with `ENEL585_F26_ProjectOrientation_PLECS_20260909`.

Table 1: Orientation model fields

Field	Default	Use
`_P.enableswitch_SwitchState`	0	Set to 1 to start switching
`_P.Vout_ref_Value`	10	Output voltage reference
`_B.ADC1`, `_B.ADC2`	—	Measured signals (watch only)
`_B.Saturation`	—	Controller output after the limit (watch only)

In the project models the prefix is `Controller` (`Controller_P`, `Controller_B`). Your own references and gains appear there once you make them tunable (Section 4).

Table 2: Project model fields (B, D and E templates)

Field	Default	Use
`Controller_P.enableswitch_SwitchState`	0	Set to 1 to enable the PWM
`Controller_B.ADC_A`, `Controller_B.ADC_B`	—	Measured signals in physical units (watch only); channel order as in the Controller's ADC blocks

4. Make Your Own Gains Tunable

Only blocks in the Exceptions list are tunable. To tune another block, such as your compensator, add it to the list on **Coder options → Parameter Inlining**, rebuild, and check `<name>.h` for its new fields.

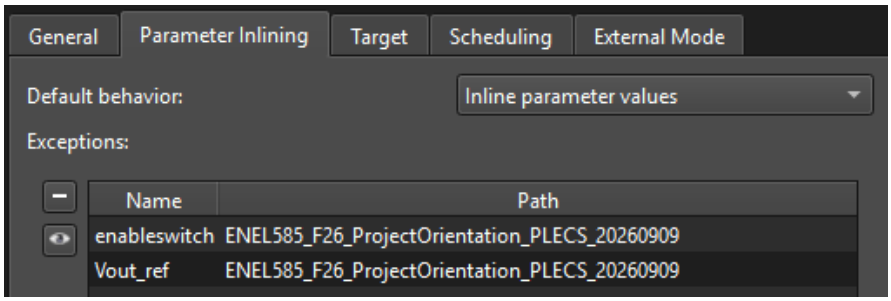


Fig. 4 – The orientation model's exceptions: only `enableswitch` and `Vout_ref` are tunable.

Troubleshooting

Problem	Fix
Load Program greyed out	Select C28xx_CPU1; connect it if needed
LaunchPad not found	Close UniFlash and any other CCS window
Stops right after Continue	Remove all breakpoints
Field missing from Watch	Not tunable or misspelled; check `<name>.h`
Edit refused while running	Pause, edit, Continue (HIL only)
Tab says source file `<name>_main.c` was not found	Harmless: the source is not on the HIL computer. Close the tab.
CCS offers to install updates	Click Cancel

Later: On the Power Bench

Never press Pause in CCS while the LaunchPad controls the real converter. Pause stops the controller, but the converter keeps running without control. Pausing is harmless only on the HIL.